Lecture 23 — RREF, pivots, free variables, and software

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Boyd video (optional): Lecture 31 — solving linear equations · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: RREF and software solvers; Lachniet Ch. 2 (linear-system commands). Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Interpret reduced row echelon form.
2. Identify pivot and free variables.
3. Describe a solution set parametrically.
4. Distinguish inconsistent, unique, and infinite-solution cases.
5. Use software to solve or row-reduce a system and verify the answer using a residual.

Warm-up

The RREF of an augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Which variables are pivots? Which are free? Write the general solution.

Fill in

Reduced row echelon form (RREF) is echelon form with these extras:

- Each pivot is _____.
- Each pivot is the _____ nonzero in its column (zeros above and below).

A **pivot variable** (basic variable) corresponds to a pivot column. A **free variable** has _____ pivot in its column; assign a parameter (e.g. $x_3 = t$).

Reading the RREF of $[A \mid b]$:

Pattern	Conclusion
A row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$	_____
Every column of A has a pivot, and consistent	_____ solution
Consistent, and at least one free variable	_____ solutions

Software. Use a solver or RREF command (Octave `rref`, Python SymPy `rref`, or a course helper). Always **verify** a candidate x by computing the residual $Ax - b$ (should be near _____).

Worked in class

W1. Classify from RREF (no original system needed).

(a)
$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right]$$

(b)
$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right]$$

(c)
$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

W2. Parametric form. From
$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right],$$
identify free variable(s) and write $x = (x_1, x_2, x_3)$ in terms of parameter(s).

W3. Software + residual. For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 11 \end{bmatrix},$$

solve $Ax = b$ with Octave or Python. Compute $Ax - b$ and confirm it is (numerically) zero.

Octave sketch: `A\[b]` or `rref([A b])` then read x.

Python sketch: `np.linalg.solve(A,b)` for square invertible A, or SymPy `Matrix.hstack(A,b).rref()`.

W4. Inconsistent check. Invent a 2×2 augmented RREF that is inconsistent. What would $Ax - b$ look like if software "forced" a least-squares answer? (One sentence — foreshadow Week 15.)

Your turn

Y1. From RREF

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

list pivot variables, free variables, and the general solution.

Y2. True or false: if RREF has a row of all zeros including the augmented entry, the system is inconsistent.

Y3. You solve $Ax = b$ in software and get $x = (1, -1)$. You compute $Ax - b = (0.0,\ 10^{-15})$. Was the solve successful? Why check the residual?

Y4. A 3×5 matrix A (after RREF of $[A \mid b]$, consistent) has 3 pivots. How many free variables? Unique solution or infinitely many?

Answers (Your turn)

Y1. Pivot variables x_1, x_2; free $x_3 = t$. Then $x_1 - 2t = 5$ and $x_2 + 3t = -1$, so

$$x = (5 + 2t,\, -1 - 3t,\, t), \quad t \in \mathbb{R}.$$

Y2. **False.** A zero row including the right-hand side is consistent (redundant equation). Inconsistency needs $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$.

Y3. Yes — residual is numerically zero (roundoff). Checking $Ax - b$ catches wrong x, singular/ill-conditioned cases, and copy-paste mistakes.

Y4. $n = 5$ unknowns and 3 pivots $\Rightarrow$ 2 free variables $\Rightarrow$ infinitely many solutions (when consistent).